AI Project Delivery Plan

LectureLens AI — Real-Time Note-Taking for Students with ADHD & Hearing Impairments

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EXECUTIVE SUMMARY

LectureLens AI is a real-time AI note-taking system that converts live lectures into structured cognitive signals under a strict 3-second latency constraint. The project prioritizes accuracy, safety, and real-time performance over conversational richness. Success depends on meeting strict evaluation thresholds — 90% recall for Action Items — and maintaining a closed-loop HITL improvement system. The TPM owns cross-functional delivery, risk control, and go/no-go execution for pilot launch across 2 universities with 20 ADHD STEM students.

1. PROJECT OVERVIEW

LectureLens AI is a browser-based real-time lecture companion for students with ADHD or hearing impairments. It transcribes live audio and uses an AI agent to extract Action Items and Key Concepts, surfacing them in a live sidebar — without requiring any student interaction during the lecture.

Delivery objective: Ship a browser MVP with 3s end-to-end latency and 85% extraction accuracy, validated in a 4-week structured pilot with 20 students across 2 universities.

- Action Item: Any instruction requiring future student action — assignment, deadline, task, exam prep.
- Key Concept: Any core idea or definition central to the lecture. Does NOT require action.

2. BUSINESS PROBLEM & PROJECT JUSTIFICATION

Students with ADHD and hearing impairments cannot simultaneously listen and write notes. Existing tools output raw unstructured text — no real-time signal distinguishes a deadline from background context.

Project justification: No existing tool provides a real-time structured extraction layer during live lectures. This is the delivery opportunity — and the constraint that drives every coordination decision in this plan.

3. TARGET USERS

Persona	Core Delivery Need
Maya, 20 — ADHD, engineering major	Auto-extracted Key Concepts — removes cognitive split-attention load
James, 22 — moderate hearing loss	Accurate live transcript + highlighted Action Items he would otherwise miss
Priya, 19 — ESL international student	Structured sidebar organising fast English speech into digestible concepts

Sam, 21 — general student, fast CS lectures	AI-distilled notes cutting post-lecture review time

4. DELIVERY CONSTRAINTS & SYSTEM OVERVIEW

Architecture detail is owned by Engineering and ML. This section covers constraints that drive coordination and planning decisions.

- **3-second latency budget:** The hardest constraint. All tooling and architecture decisions flow from it. Any scope change must be evaluated here first.
- **Real-time requirement:** Post-lecture processing has zero value. AI extraction must happen during the lecture — rules out batch processing.
- **Accuracy floor:** A missed or hallucinated deadline causes immediate churn. Accuracy is a delivery risk — must be in QA scope before any go/no-go decision.
- **Tooling selections:** All tool and model choices satisfy the latency and accuracy constraints above. Decisions are reversible and owned by Engineering and ML. TPM tracks delivery impact only.

5. DECISION LOG

Documents key decisions, options considered, and rationale. Maintained by TPM throughout project lifecycle.

Decision	Options Considered	Final Choice	Rationale
LLM model tier	GPT-4, Claude Sonnet, GPT-4o-mini	GPT-4o-mini	Only option meeting 3s latency budget; GPT-4/Sonnet inference alone exceeds full budget
STT engine	Whisper, Google STT, AssemblyAI	Whisper Large v3	Best WER on academic audio; self-hostable at scale for cost control
Processing architecture	Batch (post-lecture) vs streaming (real-time)	Streaming	Batch has zero value for real-time use case — rejected at project scoping
Storage for transcripts	pgvector (open-source) vs Pinecone	pgvector for MVP; Pinecone at scale	Cost efficiency at MVP scale; migration path defined for growth phase
Sidebar delivery model	Sidebar vs chat interface vs inline	Passive sidebar	ADHD users cannot context-switch mid-lecture; sidebar requires zero interaction

6. SUCCESS METRICS HIERARCHY

Metrics are tiered by strategic importance. North Star drives all trade-off decisions. Supporting metrics diagnose, not define, success.

North Star Metric: Reduction in manual note-taking effort reported by students — target 50% reduction within 2 weeks. This is the only metric that defines product success.

Product Metrics (measure core AI quality):

- Action Item recall: 90% on explicit deadline signals — CRITICAL threshold
- Key Concept precision: 85% confirmed correct by human review
- F1 Score (combined): 87% across both extraction categories

System Health Metrics (measure delivery reliability):

- P95 latency: 3 seconds — hard ceiling
- System uptime: 99.5% during academic hours (8am–10pm)
- STT Word Error Rate: 10% on academic English

Operational Metrics (measure continuous improvement):

- Drift detection time: 48 hours from signal to reviewed response
- Eval coverage: 100% of releases tested against fixed eval set before deployment
- HITL feedback rate: 25% of sessions — signals feedback loop health

7. ACCEPTANCE CRITERIA & THRESHOLDS

All metrics validated against a fixed evaluation dataset. Reviewed weekly by PM. No metric waived without explicit stakeholder sign-off.

Metric	Threshold	Owner
Word Error Rate (STT)	10% on academic English	ML
Precision — Key Concepts	85% confirmed correct by human review	ML + PM
Recall — Action Items (CRITICAL)	90% on explicit Action Items	ML + PM
F1 Score (combined)	87% across both categories	ML
Latency P50 / P95 / P99	1.5s / 3.0s / 6.0s	Engineering
Confidence gate	Items below 0.65 held pending until next chunk confirms	Engineering + ML
System uptime	99.5% during academic hours	Engineering

Pilot North Star	50% of students report 50% reduction in note-taking effort within 2 weeks	PM

8. GO / NO-GO CRITERIA

TPM owns go/no-go decision at end of Phase 3. Binary — no partial go-live.

NO-GO if ANY of the following:
- P95 latency > 3 seconds in eval set testing
- Action Item recall < 90% on explicit deadline signals
- Any CRITICAL hallucination observed in the evaluation set
- University compliance sign-off not received
- Evaluation dataset incomplete or not reviewed
- Monitoring dashboards not live before first user session

GO requires ALL of the following:
- All Section 7 acceptance criteria pass on fixed eval set
- Zero CRITICAL failure modes in QA staging
- Legal and IT sign-off received for FERPA compliance and consent flow
- Pilot user onboarding confirmed at both universities
- Rollback mechanism tested and verified

9. PROJECT OUTCOME DEFINITION

Binary clarity for executive reporting and stakeholder alignment.

SUCCESS = ALL conditions met:
- Pilot launches on time (Week 4)
- P95 latency 3s achieved and sustained
- Action Item recall 90% in eval set
- 50% of pilot users report 50% reduction in note-taking effort by Week 4
- Zero CRITICAL hallucination failures during pilot

FAILURE = ANY condition below:
- CRITICAL hallucination observed during pilot (missed or fabricated deadline)
- Latency consistently exceeds 3s — degraded mode becomes the default experience
- Pilot adoption below 50% of enrolled users by Week 2
- University compliance approval withdrawn

10. NON-FUNCTIONAL REQUIREMENTS

Requirement	Specification	Owner
Latency	P50 1.5s, P95 3s, P99 6s. Real-time = 3s perceived lag.	Engineering

Scalability	10,000 concurrent sessions at launch; auto-scale required	Engineering
Reliability	99.5% uptime. Transcript continues if AI sidebar fails — graceful degradation required.	Engineering
Security	TLS 1.3 in transit. Audio not stored beyond active session. FERPA-compliant.	Legal / IT
Privacy	No PII beyond email + session metadata. Audio deleted after transcription. GDPR + CCPA.	Legal / IT
Accessibility	WCAG 2.1 AA — high contrast, keyboard navigation, screen reader compatible.	Product / Design
Model updates	All model changes A/B tested on 10% of traffic before full rollout.	ML + Engineering

11. ASSUMPTIONS & CONSTRAINTS

Assumptions (validated during pilot — failure triggers scope or timeline revision):

- Students have stable internet connectivity during lectures
- Lectures are single-speaker dominant — V1 assumption; multi-speaker deferred to V3
- STT API availability is stable during pilot phase
- Student browsers support required audio and real-time delivery APIs
- Universities permit passive audio processing with explicit student consent
- Lecture audio is predominantly English academic speech

Constraints (fixed — require stakeholder approval to change):

- 3-second latency is a hard ceiling — not a target
- Audio must never be stored server-side beyond the active session
- MVP scope is browser-only — no mobile, no offline mode
- Pilot must launch at semester start — mid-semester launch kills adoption
- All model changes require A/B testing before production rollout

12. ASSUMPTION VALIDATION MATRIX

Each assumption has a defined validation method and failure impact. TPM tracks validation status weekly.

Assumption	Validation Method	Failure Impact	Owner
Stable campus WiFi during lectures	Pilot telemetry logs — latency distribution per session	High latency spikes; degraded mode becomes default	Engineering + PM
Single-speaker lecture audio dominant	Audio segmentation analysis in Phase 1 testing	Multi-speaker drift increases STT WER —	ML

		blocks accuracy threshold	
STT API stability during peak usage	Load testing in staging before pilot launch	System downtime during lectures — no recovery path in V1	Engineering
Browser API support on student devices	Compatibility testing across Chrome, Firefox, Safari	Subset of users cannot access product — reduces pilot validity	Engineering
Student willingness to provide HITL feedback	Feedback rate tracked from Session 1 (target: 25% of sessions)	HITL loop degrades; model improvement slows	PM

13. RISK REGISTER

Maintained by TPM. Reviewed weekly. Single accountable owner per risk.

Risk	Likelihood	Mitigation	Owner
Audio quality degradation in noisy lecture halls	Medium	UI guidance on mic placement; hardware partnership in roadmap	PM + Engineering
Latency exceeds 3s under poor campus WiFi	Medium	CDN-edge STT deployment; degraded mode UX with clear user notification	Engineering
LLM hallucinates Action Items not spoken	Medium	Confidence threshold (0.65); HITL flags; daily eval set as primary gate	ML + PM
University recording policy restrictions	Low-Medium	Consent flow built in; pilot at semester start only; policy guidance page	PM + Legal
Low pilot adoption rate	Medium	Adoption risk mitigated via workflow integration; weekly check-ins with cohort	PM
Model drift post-launch	Medium	Weekly eval + HITL loop detects drift within 48 hours; rollback if precision drops 5%	ML + PM
Eval dataset not ready before Phase 3	Medium	Annotation starts Phase 1 as parallel workstream; TPM tracks as critical path item	PM + ML

Student over-reliance on AI without verification	Low	Full transcript visible; confidence indicator shown; calibration in Session 1	Product

14. DEPENDENCY RISK HEATMAP

Risk intensity by dependency — used to prioritize TPM attention and escalation during execution.

Risk Level	Dependency	If Delayed
HIGH	STT latency stability	Blocks Phase 2 entirely — no extraction layer can be built on unstable latency foundation
HIGH	LLM structured output reliability	Blocks Phase 3 eval set testing — JSON schema failures cascade into sidebar and QA
MEDIUM	Evaluation dataset readiness	Blocks Phase 3 gate — annotation must start Phase 1 as parallel workstream
MEDIUM	University compliance sign-off	Blocks pilot launch — cannot onboard users without legal approval
MEDIUM	Real-time delivery stability	Blocks degraded mode design and testing — frontend dependency
LOW	Export feature (PDF/Markdown)	Does not block pilot launch — post-session feature, can ship in patch
LOW	Consent banner implementation	Low complexity; not on critical path

15. USER BEHAVIOR RISKS

AI TPM responsibility extends beyond system delivery to behavioral system design. These risks are distinct from technical failure modes.

Behavior Risk	Trigger	Mitigation
Over-trust in AI suggestions	Student acts on sidebar item without verifying against lecture context	Confidence indicator shown per item; verbatim transcript phrase displayed as evidence anchor
Reduced active listening	Student delegates all attention to sidebar, disengages from lecture entirely	Full transcript kept visible alongside sidebar; UI design keeps lecture as primary focus
Cognitive overload from sidebar	Too many items appear simultaneously — sidebar becomes noise, not signal	Rate limiting (max 3 items per 30s window); semantic deduplication; Action Items always prioritized

Trust collapse after single error	One hallucinated assignment causes student to abandon product permanently	Onboarding calibration in Session 1; easy correction flow; expectation-setting on AI confidence

16. FAILURE MODES & MITIGATION

CRITICAL = cannot ship. HIGH = recoverable. MEDIUM = tolerable if under 5% of sessions. Zero CRITICAL failures required at go/no-go gate.

Severity	Failure Type	Mitigation
CRITICAL	Hallucination — agent tags assignment not spoken	Confidence threshold + HITL flag + weekly hallucination audits
CRITICAL	Miss — agent fails to tag explicit deadline	90% recall target; deadline signals in system prompt; daily eval set
HIGH	STT timeout — transcript stops updating	Disable AI sidebar first. Retry with backoff. Fallback to basic browser transcription labeled clearly.
HIGH	Real-time delivery drop mid-lecture	Auto-reconnect; session state preserved; no data lost on recovery
HIGH	Full connectivity loss	Audio buffered locally; replayed automatically on reconnect
MEDIUM	Sidebar clutter	Max 3 items per 30s window; semantic deduplication
MEDIUM	Low user trust in early sessions	Confidence indicator; easy correction flow; calibration in Session 1

17. HITL DESIGN & CONTINUOUS IMPROVEMENT

> *HITL reduces iteration cycle from weeks to 24-48 hours via structured feedback loop and daily eval gating. This is a delivery requirement — scoped into the project plan from Day 1.*

Mechanism: Thumbs up/down on each sidebar item logs labelled pairs automatically — no manual data labelling required.

- **Daily automated eval:** Fixed test set run daily. Primary quality gate — triggers immediate review if thresholds breach.
- **Weekly PM sampling:** 50 random live sessions reviewed. Target: 24-48 hours from signal to deployed fix.
- **Drift alert:** Precision drops 5%+ week-over-week triggers escalation to ML lead and stakeholder notification.

18. ENGINEERING VS ML OWNERSHIP BOUNDARY

Prevents blurred ownership — the primary cause of AI project delivery failure in cross-functional teams.

Engineering Owns	ML / Data Science Owns	Shared
Streaming pipeline and audio capture	Prompt design and iteration	Confidence threshold tuning
Latency optimization	Evaluation metrics and eval set maintenance	Failure mode classification
Real-time delivery and connection stability	Hallucination control and mitigation	A/B test design and analysis
System reliability and uptime	Model selection and upgrade evaluation	Go/no-go criteria validation
Degraded mode implementation	Drift monitoring and alert thresholds	Pilot quality review

19. MVP SCOPE & OUT OF SCOPE

In scope for MVP:

- Live audio transcription (streaming, 2s lag)
- AI sidebar — real-time Action Items and Key Concepts with confidence gating
- Session review mode — full transcript + extracted highlights post-lecture
- Export — PDF and Markdown
- HITL feedback buttons — thumbs up/down per sidebar item
- Consent banner at session start

Out of scope (explicitly deferred):

- Mobile app — V2
- Offline mode with local inference — V2
- Multi-speaker diarization — V3
- LMS integrations (Canvas, Blackboard) — V3
- Quiz generation — V3
- Collaborative shared notes — V4

20. PILOT PLAN & SUCCESS CRITERIA

20 ADHD STEM students, 2 universities, 10 lectures each, 200 total sessions, 4 weeks. TPM coordinates with weekly stakeholder readout.

Success Criterion	Target	Measured By
North Star: 50% reduction in note-taking effort within 2 weeks	60% confirm	PM (survey + behavioral logs)
Week-2 retention	70%	Engineering (session logs)
Action Item extraction accuracy	85%	ML + PM (eval set)

Latency P95	3 seconds	Engineering (monitoring)
Participants wanting to continue post-pilot	75%	PM (exit survey)
Post-lecture re-listening time reduction	50% vs. baseline	PM (behavioral logs)

21. EXECUTION PLAN & MILESTONES

TPM owns milestone tracking, phase gate decisions, and escalation throughout all phases.

- **Phase 1 — Core Pipeline (Week 1-2):** Audio capture, STT integration, basic transcript display. Eval dataset annotation starts as parallel workstream. Gate: P95 latency 3s confirmed.
- **Phase 2 — AI Extraction Layer (Week 2-3):** LLM agent integration, structured output validation, sidebar rendering, confidence gating, rate limiting.
- **Phase 3 — Evaluation & Stability (Week 3-4):** Eval set testing (50 annotated lectures). Degraded mode. Gate: zero CRITICAL failures before go/no-go.
- **Phase 4 — Pilot Launch & Iteration (Week 4+):** Deploy to 20 pilot users. Weekly metric review. Iteration: 24-48 hours from signal to fix.

Milestone	Target Date	Owner	Gate Condition
Core pipeline live	End of Week 1	Engineering	P95 latency 3s confirmed
AI sidebar MVP	End of Week 2	Engineering	Structured output stable; sidebar rendering verified
Eval set pass	End of Week 3	PM + ML	Zero CRITICAL failures in eval set
Go/no-go decision	Start of Week 4	PM + Stakeholders	All gate conditions met; legal sign-off received
Pilot launch	Week 4	PM	University onboarding complete; monitoring live
Pilot readout	End of Week 8	PM + Stakeholders	North Star reviewed; V2 scope decided

22. STAKEHOLDER OWNERSHIP & RACI

R = Responsible | A = Accountable | C = Consulted | I = Informed

Activity	PM/TPM	Engineering	ML/Data	Product	Legal/IT	Stakeholders
Project plan & milestone tracking	A/R	I	I	I	I	I

Risk register & escalation	A/R	C	C	C	C	I
Pipeline & latency delivery	I	A/R	C	I	I	I
Model performance & drift	C	C	A/R	I	I	I
Acceptance criteria definition	A	C	R	C	I	C
UX design & onboarding	C	I	I	A/R	I	I
FERPA compliance & privacy	C	I	I	I	A/R	I
Pilot coordination & onboarding	A/R	I	I	C	C	I
Go/no-go decision	A	C	C	C	C	R
Weekly stakeholder reporting	A/R	I	I	I	I	I

23. CRITICAL PATH & DEPENDENCIES

> *Critical path determines the earliest possible pilot launch. TPM owns dependency tracking and escalation. Any delay in a critical path item directly delays go-live.*

- **1. STT latency stability:** P95 3s must be confirmed in Phase 1 before extraction layer is built. Blocks Phase 2. TPM escalates to Engineering if not resolved by end of Week 1.
- **2. LLM structured output reliability:** Consistent structured output required before sidebar rendering begins. Phase 2 instability blocks Phase 3 eval testing. ML lead owns; TPM tracks daily.
- **3. Real-time delivery stability:** Frontend rendering within latency budget required before degraded mode can be designed and tested. Engineering owns; TPM monitors in Phase 2.
- **4. Evaluation dataset readiness:** Annotated eval set must be complete before Phase 3. Parallel workstream — TPM ensures annotation starts Phase 1.
- **5. University compliance sign-off:** Legal and IT approval required before any user is onboarded. TPM initiates in Phase 1. Cannot be left to week of pilot launch.

24. LAUNCH READINESS CHECKLIST

Pilot cannot start unless ALL items are checked. TPM signs off on each item individually.

Checklist Item	Owner	Status
Evaluation dataset fully annotated and reviewed	PM + ML	
All Section 7 acceptance criteria pass on fixed eval set	PM + ML	
Zero CRITICAL failures in eval run	PM + QA	
Monitoring dashboards live (latency, accuracy, drift)	Engineering	
Rollback mechanism tested and verified	Engineering	
Failure modes tested in staging environment	Engineering + ML	
Legal / FERPA approval complete	Legal / IT	
Consent flow tested end-to-end	Engineering + Legal	
Support escalation path defined and communicated	PM	
University pilot user onboarding confirmed	PM	
Degraded mode UX tested and verified	Engineering	

25. COST & SCALING MODEL

Cost model maintained by TPM in collaboration with Engineering. Reviewed monthly and at each major scope change.

Estimated cost per lecture session (MVP):

- STT (Whisper API): ~/bin/sh.006 per minute of audio — 60-min lecture = ~/bin/sh.36
- LLM inference (GPT-4o-mini): ~/bin/sh.00015 per 1K tokens — approx /bin/sh.02-0.05 per session depending on lecture density
- Storage and compute: ~/bin/sh.01 per session at MVP scale
- Estimated total: ~/bin/sh.40-0.50 per lecture session at MVP scale

Primary cost drivers:

- **STT usage:** Dominant cost at scale. Mitigation: self-host Whisper at 10K+ daily sessions — reduces STT cost by ~80%.
- **LLM inference:** Controlled via GPT-4o-mini selection and chunk aggregation — reduces API calls by approximately 30% vs naive per-chunk inference.

Scaling assumptions:

- 10,000 concurrent sessions at peak — requires auto-scaling infrastructure
- Cost per session decreases significantly with self-hosted STT at scale
- LLM cost grows linearly with usage unless batching optimization is implemented in V2

26. SCALING FAILURE MODES

What breaks first at 10,000 concurrent users — and the mitigation strategy for each.

Component	Failure at Scale	Mitigation Strategy
STT API (Whisper)	Becomes throughput bottleneck — queue delay increases, P95 latency exceeds 3s budget	Self-host Whisper at regional edge; implement request queuing with priority for active sessions
LLM inference (GPT-4o-mini)	Latency variance increases under load — P95 violation risk rises	Rate limiting per session; fallback to cached responses for common query patterns; request batching
Real-time delivery (WebSocket)	Connection congestion causes sidebar lag across high-volume sessions simultaneously	Regional WebSocket servers; connection load balancing; graceful degradation to polling fallback
Eval and monitoring system	Monitoring dashboards lag behind real-time signal at high session volume	Async monitoring pipeline; sampled eval at scale (10% of sessions) rather than 100% coverage

27. POST-LAUNCH OPERATING MODEL

TPM owns the operating cadence post-pilot. AI systems require active management — not passive monitoring.

Cadence	Activity	Owner
Daily	Automated eval set run — alerts if thresholds breach	ML (automated) + PM (review)
Daily	Latency and uptime dashboard review	Engineering
Weekly	Accuracy and drift analysis — 50 sampled sessions reviewed	PM + ML
Weekly	Stakeholder update — metrics, risks, decisions made	PM
Monthly	Model upgrade evaluation — benchmark new versions on eval set before A/B test	ML + PM
Monthly	Cost review — actual vs projected per-session cost; scaling plan update	PM + Engineering
On CRITICAL failure	Rollback within 24 hours to prior stable prompt version; incident report within 48 hours	ML + Engineering + PM

On drift alert (>5% precision drop)	Immediate ML review; prompt fix deployed within 48 hours; stakeholder notified	ML + PM

28. KEY TAKEAWAYS

> *AI systems must be managed as probabilistic systems with defined evaluation, monitoring, and rollback strategies — not deterministic software. The TPM role is to ensure these mechanisms are in place before launch, not after the first failure.*

- The 3-second latency budget is the single most important constraint — every coordination and tooling decision flows from it.
- RACI and critical path are living documents — ambiguous ownership is the primary cause of AI project delays in cross-functional teams.
- Go/no-go criteria must be explicit and binary — removes subjectivity from the most important delivery decision.
- Assumptions must be documented and validated during pilot — failure of any assumption triggers a formal scope or timeline revision.
- HITL is a delivery mechanism — reduces iteration time from weeks to 24-48 hours and must be in scope from Day 1.
- Post-launch operating model is not optional — AI systems degrade without active management; the TPM owns the cadence.